

PKU-Zurich Summer School

Lecture-1: Continuous Time Methods

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Outline

Introduction

Itô Processes

Kolmogorov Forward Equation (KFE)

Optimal Control

Brownian Shocks

Poisson Shocks

Introduction

Stochastic Processes

- I assume some familiarity with the following concepts:
 - ▶ Random variables, probability distributions
 - ▶ Expectation, variance, covariance, correlation
 - ▶ (Not required but helpful) σ -Algebras, Filtrations, Norms, Convergence concepts — e.g. convergence in probability, convergence in distribution, convergence in L^p

A good *accessible* reference for Stochastic Calculus in Finance is: *Shreve* (2004).

Advanced references: *Karatzas and Shreve* (1998), *Øksendal* (2003), *Applebaum* (2009)

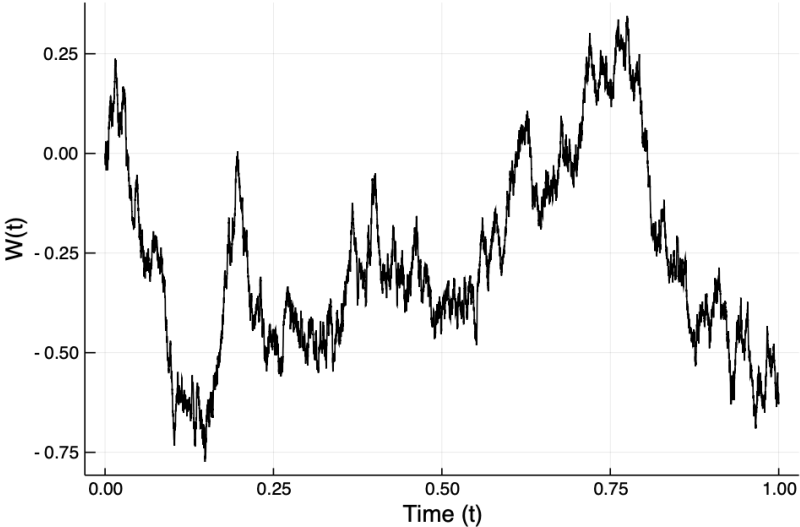
★ This material is based on lecture notes by Jonathan Payne.

Brownian Motion

Definition: A k -dimensional Brownian motion is a stochastic process $\{W_t\}_{t \geq 0}$ such that:

1. $W_0 = 0$ a.s.
2. (Normal increments): $\forall x, t \in \mathcal{T}$ for which $s \leq t$, the difference $W_t - W_s \sim N(0, (t - s)I_k)$
3. (Independent increments): $\forall t_1, t_2, \dots, t_n \in \mathcal{T}$ with $0 \leq t_1 < t_2 < \dots < t_n$, the random variables $W_{t_1}, W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}}$ are independent
4. (Continuous sample paths): W_t is continuous for almost all $\omega \in \Omega$

Brownian Motion



Brownian Motion

More properties of Brownian motion

- BM has the Markov property: $P(W_t \leq x | W_s) = P(W_t \leq x | W_s, W_u)$ for all $0 \leq s \leq u \leq t$
- BM is a martingale: $E[W_t | \mathcal{F}_s] = W_s$ for all $0 \leq s \leq t$
- BM has infinite variation on any finite interval $[0, t]$ for any $t > 0$

$$\langle W, W \rangle_t^1(\omega) = \infty, \forall t > 0$$

$$[W, W]_t(\omega) = \langle W, W \rangle_t^2(\omega) = t, \forall t \geq 0$$

- Almost every sample path of Brownian motion is not differentiable at any point in time.

Poisson Process

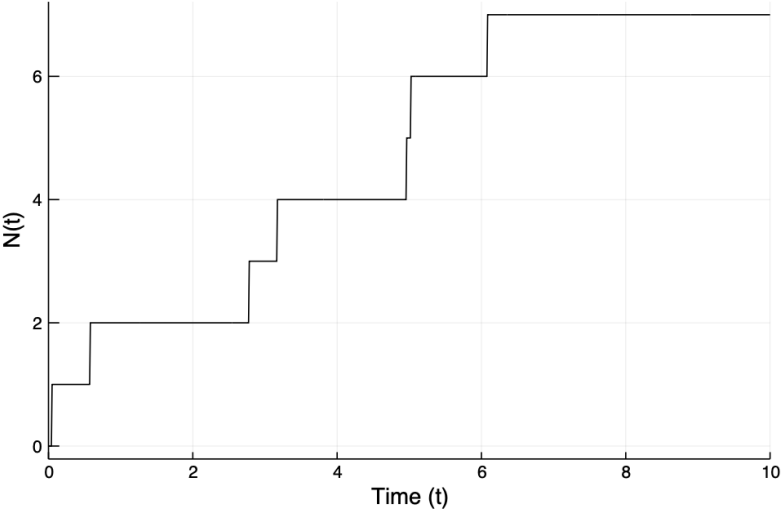
Definition: A (simple) Poisson process with parameter $\lambda > 0$ is a stochastic process $\{N_t\}_{t \geq 0}$ such that:

1. $N_0 = 0$ a.s.
2. (Stationary Poisson increments): If $s < t$, then $N_t - N_s \sim \text{Pn}(\lambda(t - s))$. That is,

$$\mathbb{P}(N_t - N_s = k) = e^{-\lambda(t-s)} \frac{(\lambda(t-s))^k}{k!}$$

3. (Independent increments): $P_t - P_s$ is independent of $\mathcal{F}_s$ for any $0 \leq s < t \leq T$.
4. (Right continuous sample paths): P_t is right continuous for all $t \geq 0$.

Poisson Process



Poisson Process

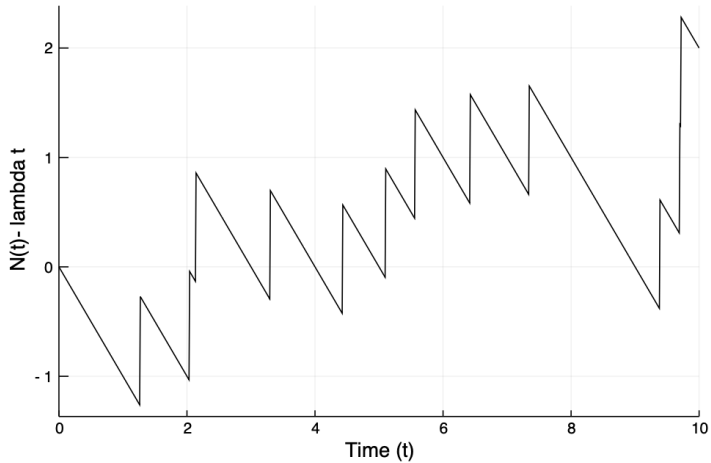


Figure: Sample path for the compensated Poisson process $\bar{N}_t = N_t - \lambda t$.

Poisson Process

More properties of Poisson process

- For any given k , the random inter-jump time $\tau_k := T_{k+1} - T_k$ is exponentially distributed with parameter λ .
- The Poisson process has the Markov property.
- The Poisson process is *not* a Martingale. The *compensated* Poisson process, $N_t - \lambda t$ is a martingale.
- The Poisson process has finite variation and quadratic variation

$$\langle N, N \rangle_t^1(\omega) = \infty, \forall t > 0$$

$$[N, N]_t(\omega) = \langle N, N \rangle_t^2(\omega) = t, \forall t \geq 0$$

As a consequence, we have that

$$[W, N]_t(\omega) = \langle W, N \rangle_t^2(\omega) = 0, \forall t \geq 0$$

Itô Processes

Itô Processes

Definition: An Itô process is a stochastic process $\{X_t\}_{t \geq 0}$ such that:

1. $X_0 = x$ a.s.
2. (Itô process): $X_t = X_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s + \int_0^t \alpha(s, X_{s-}) dN_s$

Where $b(s, x), \sigma(s, x), \alpha(s, x, u)$ are functions of s, x, u and W is a Brownian motion and N is a Poisson process. Ito process X has the stochastic differential on $[0, T]$:

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t + \int_0^t \alpha(t, X_t, u)dN_u$$

Itô Processes

- If $dX_t = \mu_t dt + \sigma_t dW_t$, then:

$$[X, X]_t = \int_0^t \sigma_s^2 ds$$
$$\implies d[X, X]_t = \sigma_t^2 dt$$

- If $dX_t = \mu_t dt + \sigma_t dW_t + \gamma_t dN_t$, then:

$$[X, X]_t = \int_0^t \sigma_s^2 ds + \int_0^t \gamma_s^2 dN_s$$
$$\implies d[X, X]_t = \sigma_t^2 dt + \gamma_t^2 dN_t$$

Itô's Lemma

Lemma 1 (Univariate case). *If $f(t, x)$ is a twice continuously differentiable function, and X_t is an Itô process of the form $dX_t = \mu_t dt + \sigma_t dW_t$, then:*

$$df(t, X_t) = \frac{\partial f}{\partial t}(t, X_t)dt + \frac{\partial f}{\partial x}(t, X_t)dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t)d[X, X]_t$$

Itô's Lemma

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Lemma 2 (Multivariate case). *If $f(t, x_1, x_2, \dots, x_n)$ is a twice continuously differentiable function, and $X_t^1, X_t^2, \dots, X_t^n$ are Itô processes of the form $dX_t^i = \mu_t^i dt + \sigma_t^i dW_t + \gamma_t^i dN_t$, then:*

$$\begin{aligned} df(t, X_t^1, X_t^2, \dots, X_t^n) &= \frac{\partial f}{\partial t}(t, X_t^1, X_t^2, \dots, X_t^n)dt + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(t, X_t^1, X_t^2, \dots, X_t^n)dX_t^i \\ &\quad + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j}(t, X_t^1, X_t^2, \dots, X_t^n)d[X_t^i, X_t^j]_t \end{aligned}$$

Itô's Lemma (with jumps)

Lemma 3 (Univariate case, With jumps). *If $f(t, x)$ is a twice continuously differentiable function, and X_t is an Itô process of the form $dX_t = \mu_t dt + \sigma_t dW_t + \gamma_t dN_t$, then:*

$$df(t, X_t) = \frac{\partial f}{\partial t}(t, X_t)dt + \frac{\partial f}{\partial x}(t, X_t)dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t)d[X, X]_t + \frac{\partial f}{\partial x}(t, X_t)\gamma_t dN_t$$

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$$df(t, X_t) = \frac{\partial f}{\partial t}(t, X_t)dt + \frac{\partial f}{\partial x}(t, X_t)dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t)d[X, X]_t + \frac{\partial f}{\partial x}(t, X_t)\gamma_t dN_t$$

Lemma 4 (Two variable case, With jumps). *If $f(t, x, y)$ is a twice continuously differentiable function, and X_t and Y_t are Itô processes of the form $dX_t = \mu_t dt + \sigma_t dW_t + \gamma_t dN_t$ and $dY_t = \nu_t dt + \rho_t dW_t + \delta_t dN_t$, then:*

$$\begin{aligned} df(t, X_t, Y_t) = & \frac{\partial f}{\partial t}(t, X_t, Y_t)dt + \frac{\partial f}{\partial x}(t, X_t, Y_t)dX_t + \frac{\partial f}{\partial y}(t, X_t, Y_t)dY_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t, Y_t)d[X, X]_t \\ & + \frac{1}{2} \frac{\partial^2 f}{\partial y^2}(t, X_t, Y_t)d[Y, Y]_t + \frac{\partial^2 f}{\partial x \partial y}(t, X_t, Y_t)d[X, Y]_t + \frac{\partial f}{\partial x}(t, X_t, Y_t)\gamma_t dN_t + \frac{\partial f}{\partial y}(t, X_t, Y_t)\delta_t dN_t \end{aligned}$$

Itô's Lemma (with jumps)

Lemma 5 (Univariate case, With jumps). *If $f(t, x)$ is a twice continuously differentiable function, and X_t is an Itô process of the form $dX_t = \mu_t dt + \sigma_t dW_t + \gamma_t dN_t$, then:*

$$df(t, X_t) = \frac{\partial f}{\partial t}(t, X_t)dt + \frac{\partial f}{\partial x}(t, X_t)dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t)d[X, X]_t + \frac{\partial f}{\partial x}(t, X_t)\gamma_t dN_t$$

Itô's Lemma (with jumps)

Lemma 5 (Univariate case, With jumps). *If $f(t, x)$ is a twice continuously differentiable function, and X_t is an Itô process of the form $dX_t = \mu_t dt + \sigma_t dW_t + \gamma_t dN_t$, then:*

$$df(t, X_t) = \frac{\partial f}{\partial t}(t, X_t)dt + \frac{\partial f}{\partial x}(t, X_t)dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t)d[X, X]_t + \frac{\partial f}{\partial x}(t, X_t)\gamma_t dN_t$$

Lemma 6 (Multivariate case, With jumps). *If $f(t, x_1, x_2, \dots, x_n)$ is a twice continuously differentiable function, and $X_t^1, X_t^2, \dots, X_t^n$ are Itô processes of the form $dX_t^i = \mu_t^i dt + \sigma_t^i dW_t + \gamma_t^i dN_t$, then:*

$$\begin{aligned} df(t, X_t^1, X_t^2, \dots, X_t^n) &= \frac{\partial f}{\partial t}(t, X_t^1, X_t^2, \dots, X_t^n)dt + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(t, X_t^1, X_t^2, \dots, X_t^n)dX_t^i \\ &+ \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j}(t, X_t^1, X_t^2, \dots, X_t^n)d[X_t^i, X_t^j]_t + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(t, X_t^1, X_t^2, \dots, X_t^n)\gamma_t^i dN_t \end{aligned}$$

Algebra rules

	dt	$dW(t)$	$dN(t)$
dt	0	0	0
$dW(t)$	0	dt	0
$dN(t)$	0	0	dN_t

Algebra rules (complete)

Differential Multiplication	Result
dt	dt
dW_t	dW_t
dN_t	dN_t
$(dW_t)^2$	dt
$(dW_t)^m, m \geq 3$	0
$dt dW_t$	0
$(dt)^\alpha (dW_t)^m, \alpha > 0, m \geq 1$	0
$(dN_t)^m, m \geq 1$	dN_t
$dt dN_t$	0
$(dt)^k (dN_t)^m, k \geq 1, m \geq 1$	0

Kolmogorov Forward Equation (KFE)

Kolmogorov Forward Equation (KFE)

Consider the following environment.

- Population: continuum of agents $I := [0, 1]$.
- Each agent i has a state variable $X_t^i \in \mathbb{R}$ that follows

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t^i(X_t^i)dW_t$$

where W is a Brownian motion, and $\mu_t(x)$ and $\sigma_t(x)$ are Lipschitz in x and t .

- Let $G_t = \mathcal{L}(X_t^i)$ denote the population distribution across state X_t^i .
- Let g_t denote population density across state X_t at time t .

Kolmogorov Forward Equation (KFE)

Theorem: Suppose that each agent's state X_t^i evolves according to

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t^i(X_t^i)dW_t$$

Then, the population density g_t satisfies the following partial differential equation:

$$\frac{\partial g_t}{\partial t} = -\frac{\partial}{\partial x} (\mu_t(x)g_t(x)) + \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma_t^2(x)g_t(x))$$

with the initial condition $g_0(x) = \psi(x)$.

KFE: Higher dimensions

Theorem: Suppose that each agent's state $X_t^i \in \mathbb{R}^k$ evolves according to

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t^i(X_t^i)dW_t$$

Then, the population density g_t satisfies the following partial differential equation:

$$\frac{\partial g_t}{\partial t} = - \sum_{i=1}^k \frac{\partial}{\partial x_i} (\mu_t(x)g_t(x)) + \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k \frac{\partial^2}{\partial x_i \partial x_j} (\sigma_t^2(x)g_t(x))$$

with the initial condition $g_0(x) = \psi(x)$.

KFE with aggregate shocks

Consider the following environment.

- Population: continuum of agents $I := [0, 1]$.
- Each agent i has a state variable $X_t^i \in \mathbb{R}$ that follows

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t^i(X_t^i)dW_t^i + \sigma_t^0(X_t^i)dW_t^0$$

where W^i and W^0 are Brownian motions, and $\mu_t(x)$ and $\sigma_t(x)$ are Lipschitz in x and t .

- Let $G_t = \mathcal{L}(X_t^i)$ denote the population distribution across state X_t^i .
- Let g_t denote population density across state X_t at time t .

KFE with aggregate shocks

Theorem: Suppose that each agent's state $X_t^i \in \mathbb{R}^k$ evolves according to

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t^i(X_t^i)dW_t^i + \sigma_t^0(X_t^i)dW_t^0$$

Then, the population density g_t satisfies the following partial differential equation:

$$\begin{aligned} \frac{\partial g_t}{\partial t} = & -\partial_x[\mu_t(x)g_t(x)]dt - \partial_x[\sigma_t^0(x)g_t(x)]dW_t^0 \\ & + \frac{1}{2}\sigma_{xx}[(\sigma_t^2(x) + (\sigma_t^0)^2)g_t(x)]dt \end{aligned}$$

with the initial condition $g_0(x) = \psi(x)$.

Birth and death

Consider the following environment.

- Population: continuum of agents $I := [0, 1]$.
- Each agent i has a state variable $X_t^i \in \mathbb{R}$ that follows

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t^i(X_t^i)dW_t$$

where W is a Brownian motion, and $\mu_t(x)$ and $\sigma_t(x)$ are Lipschitz in x and t .

- Agents killed according to a Poisson process with intensity $\lambda_k(X_t)$.
- Agents reborn at y according to Poisson process with intensity $\lambda_b(X_t, y)$.

Suppose $\lambda_k(x)$ and $\lambda_b(x, y)$ are continuously distributed. The KFE then becomes

$$\frac{\partial g_t}{\partial t} = -\partial_x[\mu_t(x)g_t(x)]dt - \lambda_k(x)g_t(x)dt + \int_0^1 \lambda_b(x, y)g_t(y)dydt$$

Optimal Control

Optimal Control

Multiple approaches

1. Dynamic Programming
(Easiest to solve but requires recursivity)
2. Forward-Backward Stochastic Differential Equations (FBSDE)
(Required when process cannot be expressed recursively)
3. Martingale Methods
(Useful when there is obvious SDF for pricing)

Applications: Consumption-Savings, Portfolio Choice, Life-Cycle, etc.

References: *Björk* (2012), *Pham* (2009)

4. Optimal Control

4.1. Brownian Shocks

HJB

Consider the following setup.

- There is an agent with a state variable $X_t \in \mathbb{R}$ that follows

$$dX_t = \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t$$

where W is a Brownian motion, and $\mu_t(x, c)$ and $\sigma_t(x, c)$ are Lipschitz in x and c and t .

- The agent's value function is

$$V(x) = \sup_{c_t \in \mathcal{C}(x)} \mathbb{E} \left[\int_0^T e^{-\rho s} u(C_s) ds \right]$$

where ρ is the discount rate and $u(x)$ is a utility function.

- The agent's infinite horizon stochastic optimal control problem is to find the optimal control c_t subject to the dynamic constraint $dX_t = \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t$ and the initial condition $X_0 = x$.

HJB (Infinite Horizon)

Proposition 1 (Necessity of the HJBE). *Suppose that X follows*

$$dX_t = \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t$$

and suppose that $X_t = x$ and value function is defined as

$$V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \mathbb{E} \left[\int_t^T e^{-\rho s} u(C_s) ds \right]$$

then the HJBE is given by

$$\rho V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \left(u(c_t) + \frac{\partial V_t}{\partial x} \mu_t(x, c_t) + \frac{1}{2} \frac{\partial^2 V_t}{\partial x^2} \sigma_t^2(x, c_t) \right)$$

Proof

There is a verification theorem for sufficiency (see *Pham (2009)* for details).

HJB (Finite Horizon)

Same set up as before except that time $t \in [0, T]$.

- For the initial condition $X_0 = x$, the value function is defined as

$$V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \mathbb{E} \left[\int_t^T e^{-\rho s} u(C_s) ds + e^{-\rho T} \psi(X_T) \right]$$

where $\psi(x)$ is a terminal payoff.

- The agent's finite horizon stochastic optimal control problem is to find the optimal control c_t subject to the dynamic constraint $dX_t = \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t$ and the initial condition $X_0 = x$.

HJB (Finite Horizon)

Proposition 2 (Necessity of the HJBE). *Suppose that X follows*

$$dX_t = \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t$$

and suppose that $X_t = x$ and value function is defined as

$$V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \mathbb{E} \left[\int_t^T e^{-\rho s} u(C_s) ds + e^{-\rho T} \psi(X_T) \right]$$

then the HJBE is given by

$$\rho V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \left(u(c_t) + \frac{\partial V_t}{\partial t} + \frac{\partial V_t}{\partial x} \mu_t(x, c_t) + \frac{1}{2} \frac{\partial^2 V_t}{\partial x^2} \sigma_t^2(x, c_t) \right)$$

As before, there is a verification theorem for sufficiency (see *Pham* (2009) for details).

Multivariate HJB (Infinite Horizon)

Same set up as before except that $X_t \in \mathbb{R}^k$.

- Suppose $X_t = (X_t^1, X_t^2, \dots, X_t^m)$ follows

$$dX_t = \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t$$

where $W = (W_t^1, W_t^2, \dots, W_t^m)$ is a Brownian motion.

- Define $\Sigma_t = \sigma_t(X_t, c_t)\sigma_t(X_t, c_t)^T$ where $\Sigma_{t,jk}$ refers to the (j, k) -th element of Σ_t .

Multivariate HJB (Infinite Horizon)

Proposition 3 (Necessity of the HJBE). *Suppose that X follows*

$$dX_t = \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t$$

and suppose that $X_t = x$ and value function is defined as

$$V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \mathbb{E} \left[\int_t^T e^{-\rho s} u(C_s) ds + e^{-\rho T} \psi(X_T) \right]$$

then the HJBE is given by

$$\rho V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \left(u(c_t) + \frac{\partial V_t}{\partial t} + \frac{\partial V_t}{\partial x} \mu_t(x, c_t) + \frac{1}{2} \frac{\partial^2 V_t}{\partial x^2} \sigma_t^2(x, c_t) \right)$$

Multivariate HJB (Finite Horizon)

Same set up as before except that time $t \in [0, T]$.

- For the initial condition $X_0 = x$, the value function is defined as

$$V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \mathbb{E} \left[\int_t^T e^{-\rho s} u(C_s) ds + e^{-\rho T} \psi(X_T) \right]$$

where $\psi(x)$ is a terminal payoff.

- The agent's finite horizon stochastic optimal control problem is to find the optimal control c_t subject to the dynamic constraint $dX_t = \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t$ and the initial condition $X_0 = x$.

Multivariate HJB (Finite Horizon)

Proposition 4 (Necessity of the HJBE). *Suppose that X follows*

$$dX_t = \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t$$

and suppose that $X_t = x$ and value function is defined as

$$V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \mathbb{E} \left[\int_t^T e^{-\rho s} u(C_s) ds + e^{-\rho T} \psi(X_T) \right]$$

then the HJBE is given by

$$\rho V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \left(u(c_t) + \frac{\partial V_t}{\partial t} + \frac{\partial V_t}{\partial x} \mu_t(x, c_t) + \frac{1}{2} \frac{\partial^2 V_t}{\partial x^2} \sigma_t^2(x, c_t) \right)$$

4. Optimal Control

4.2. Poisson Shocks

HJBE (Infinite Horizon)

Suppose that $X_t \in \mathbb{R}$ follows

$$dX_t = \mu_t(X_t, c_t)dt + h_t(X_t, c_t)dN_t$$

where N_t is a Poisson process with intensity $\lambda(x)$, and $\mu_t(x, c_t)$ and $h_t(x, c_t)$ are the drift and the jump size.

Proposition 5 (Necessity of the HJBE). *Suppose the value function is defined as*

$$V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \mathbb{E} \left[\int_t^T e^{-\rho s} u(C_s) ds + e^{-\rho T} \psi(X_T) \right]$$

then the HJBE is given by

$$\rho V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \left(u(c_t) + \frac{\partial V_t}{\partial t} + \frac{\partial V_t}{\partial x} \mu_t(x, c_t) + \lambda(x) (V_t(X_t + h_t(X_t)) - V_t(X_t)) + \frac{\partial V_t}{\partial t} \right)$$

Multivariate HJBE (Infinite Horizon)

Suppose that $X_t \in \mathbb{R}^n$ follows

$$dX_t = \mu_t(X_t, c_t)dt + h_t(X_t, c_t)dN_t$$

where N_t is an n -dim Poisson process with intensity $\lambda_k(x)$, and $\mu_t(x, c_t)$ and $h_t(x, c_t)$ are the drift and the jump size.

Proposition 6 (Necessity of the HJBE). *Suppose the value function is defined as*

$$V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \mathbb{E} \left[\int_t^T e^{-\rho s} u(C_s) ds + e^{-\rho T} \psi(X_T) \right]$$

then the HJBE is given by

$$\rho V_t(x) = \sup_{c_t \in \mathcal{C}(x)} \left(u(c_t) + \sum_{j=1}^n \frac{\partial V_t}{\partial x_j} \mu_t(x, c_t) + \sum_{k=1}^m \lambda_k(x) (V_t(X_t + h_{t,k}(X_t)) - V_t(X_t)) + \frac{\partial V_t}{\partial t} \right)$$

HJBE (Finite Horizon)

Analogous to the Brownian motion case.

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Part I

Appendix

Proof of HJB

Verification theorem details follow from the standard stochastic control argument.

See *Pham* (2009) for a complete proof. [Back](#)